

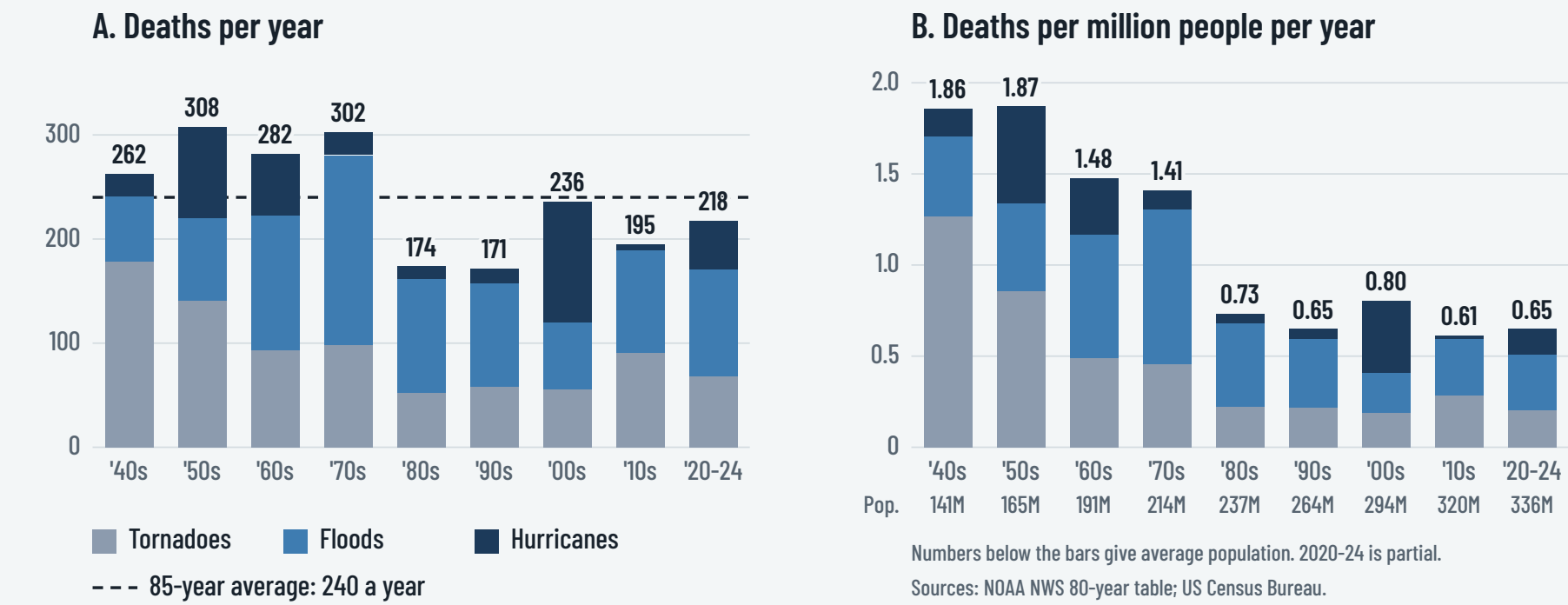
Deadlier Weather, Safer People?

Eighty-five years of US weather deaths, and why heat is harder to count

Tornadoes, floods and hurricanes kill about 240 Americans a year, close to the toll in the 1940s. The country now has more than twice as many people, so each person's risk has dropped by about two thirds. Heat is a different case. The official heat count has nearly quadrupled since the 2010s, and the ways of counting heat deaths differ drastically.

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Divide by population and the trend appears

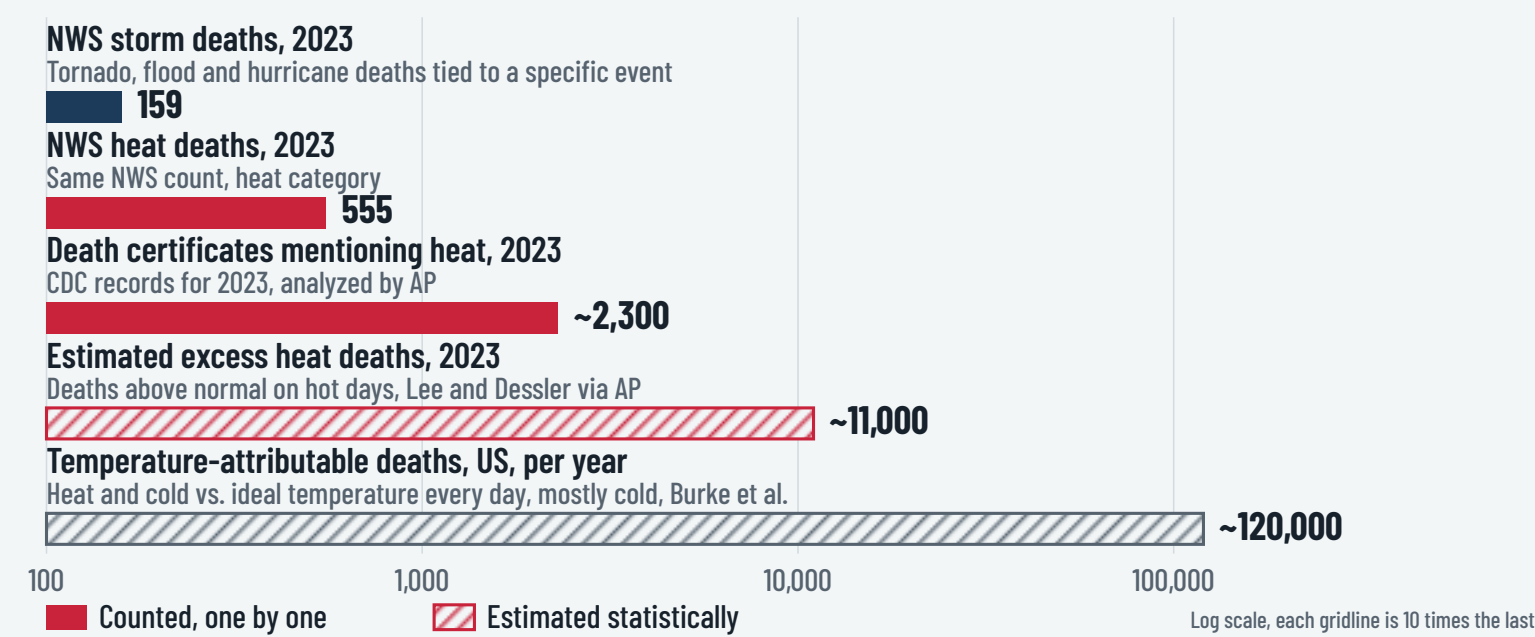


The raw counts barely move. The 1940s averaged 262 storm deaths a year, the 2010s averaged 195, and 2020 to 2024 averaged 218. Over the same period the population grew from 141 million to 336 million.

Per million Americans, the yearly death rate fell from 1.86 to 0.61.

That rate is each person's yearly odds, and it is a third of what it was. Single disasters swing the averages. Katrina caused 1,016 of the 2000s' 1,154 hurricane deaths. Without 2005, that decade averages 134 storm deaths a year, the lowest of any.

Heat deaths depend on how you count them



The National Weather Service counted about 115 heat deaths a year in the 2000s and 2010s, then 438 a year from 2020 to 2024. The NWS counts a death only when officials tie it to a specific event, so it misses most heat deaths.

Burke and coauthors use mortality records from 30 countries to estimate the deaths that would not happen if every day sat at the lowest-risk temperature. For the US that is about 5 percent of all deaths, near 120,000 a year. Cold causes at least five times as many of them as heat. Their number comes from a model, and it should never be compared with NWS counts.

The NWS logged about 30 cold deaths a year from 1995 to 2024, so adding cold barely changes the storm picture. Next to Burke's estimate, where cold dominates, that is the widest gap between counting methods here.

Before you trust a weather death number, ask

Was it counted or estimated? A list of named deaths and a statistical model measure different things.

Compared with what?

Per how many people? A rising count in a growing, aging population can still mean falling risk.

Migration and heat

Baylis and coauthors used Census county data for the 1950s to 2000s and IRS records for 1983 to 2018. Nationally, migrants chose counties that were already warm. Within a county, warming pushed people out. Cooling degree days add up how far each day's average temperature runs above 65°F. Each extra 100 cooling degree days a year cut annual net migration by 0.9 to 1.9 points.

Net domestic migration, people moving between states, Census Vintage 2025

Area	2022	2023	2025
Florida	+310,892	+183,646	+22,517
Midwest	-175,000 or worse		+16,000

July to July. Excludes births, deaths and immigration. Midwest 2022 figure covers 2021 and 2022.

Housing and insurance costs could explain part of Florida's drop. That complicates the storm story. We adapted well enough to move into heat, and now more people, and older ones, live where it is worst.

Exposure has a price

The IPCC concluded in 2012 that growing exposure of people and property, not a detected climate trend, drove the long-run rise in disaster losses. Exposure responds to prices. Subsidized flood insurance and federal disaster aid make it cheaper to stay in harm's way, so people keep building there.

Once people are exposed, prices and rules decide how much harm follows. For example, doubling home heating prices raises winter deaths by about 4 percent. California's rules requiring water, shade and breaks for outdoor workers exemplify how rules reduce harm, with the effect of heat on job injuries cut by 75 to 90 percent.

But wait

The NWS count changed. The NWS began counting heat in 1986 and logged 28 heat deaths a year through 1994. Nobody should believe that number. Better reporting from coroners as well as a generally older population are able to explain the increase.

Storm counts miss slow deaths. The NWS lists 7 hurricane deaths for 2017. Puerto Rico's official toll for Hurricane Maria, based on an excess-death study, is 2,975.

Storms are not the whole story. Burke and coauthors find the share of US deaths tied to temperature was the same in the 2010s as in the 1980s. I'm confident we got better at surviving storms, but whether we got better at dealing with extreme temperatures is still up for debate.